

Math 407 Section 1

Fall 2025

Exam I

February 12, 2025

Time Limit: 1 Hour 50 Minutes

Name (Print): _____

Student ID: _____

This exam contains 8 pages (including this cover page) and 7 problems. Check to see if any pages are missing. Enter all requested information on the top of this page, and put your initials on the top of every page, in case the pages become separated.

You may *not* use your books or notes on this exam.

You are required to show your work on each problem on this exam. The following rules apply:

- **Organize your work**, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little credit.
- **Mysterious or unsupported answers will not receive full credit.** A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations might still receive partial credit. This especially applies to limit calculations.
- If you need more space, use the back of the pages; clearly indicate when you have done this.
- If the problem asks for a proof, be sure to carefully justify your work, including any theorems from class.

Note: you may NOT use a theorem or result from class to prove something when it makes the problem entirely trivial. If you are unsure whether a particular theorem or result is allowed, just ask!

Problem	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
Total:	70	

Do not write in the table to the right.

1. (10 points) **TRUE or FALSE!** Write TRUE if the statement is true. Otherwise, write FALSE. Your response should be in ALL CAPS. No justification is required.

(a) The group $\mathbb{Q}$ with binary operation $+$ is a cyclic group.

(b) Every group of order 7 is cyclic.

(c) If G is non-Abelian and $H \leq G$ then H is also non-Abelian

(d) Every nontrivial group has a nontrivial proper subgroup.

(e) If G is a group and $x, y \in G$ then $x^3 = y^3$ if and only if $x = y$

2. (10 points)

(a) State Lagrange's Theorem

(b) Write the definition of a homomorphism from a group G to a group H .

(c) Give an example of a group of order 6 which is not Abelian.

3. (10 points) Let G be a group and $\emptyset \neq H \subseteq G$. Prove that H is a subgroup if and only if $ab^{-1} \in H$ for all $a, b \in H$.

4. (10 points) Consider the set

$$G = (\mathbb{Z}_7)^\times \times \mathbb{Z}_7 = \{(a, b) : a, b \in \mathbb{Z}_7, a \neq 0\},$$

with the binary operation

$$(a, b) * (c, d) = (a \times_7 c, b +_7 d),$$

where here $\times_7$ and $+_7$ are multiplication and addition mod 7.

- (a) Show that G has an identity element

- (b) Show each element of G has an inverse

- (c) Prove that G is a cyclic group

- (d) Explain why G is isomorphic to $\mathbb{Z}_{42}$

5. (10 points) Let H and K be subgroups of a group G .

(a) Prove that $H \cap K$ is a subgroup.

(b) Prove that if K is a normal subgroup of G , then $H \cap K$ is a normal subgroup of H .

(c) Prove that if H is also a normal subgroup of G , then $H \cap K$ is a normal subgroup of G .

6. (10 points) Consider the group $G = S_4$ and the subgroup $H = \langle (1234) \rangle$.

(a) Explicitly describe all the elements in H .

(b) Explicitly write out all of the left cosets of H in G . Carefully explain why what list are *all* the left cosets.

7. (10 points)

Suppose that G is a finite group (possibly non-Abelian). Show that if φ is a homomorphism

$$\varphi : G \rightarrow \mathbb{Z}$$

then $\varphi(x) = 0$ for all $x \in G$. In other words, the only homomorphism from G to $\mathbb{Z}$ is the trivial one.